Physicist-Engineer Nanotechnology and Quantum Applications Specialization Laboratory Measurement 2:

Ultrashort pulse processing

Abstract. Ultrashort laser pulses are important tools in many technologies involving lasers: material processing, telecommunication, laser microscopy, surgery and medical applications, etc. These pulses have high peak energy and duration at least ten times shorter than the minimum integration time of the fastest laser detectors. This imposes the need for special techniques for pulse length measurement such as autocorrelators or spectrally resolved interferometry. The frequency bandwidth of these pulses implies special techniques to maintain the temporal length during propagation through the optical system leading from the source to the target. The present laboratory exercise makes spectral and temporal measurement and dispersion compensation techniques familiar to the practicing students and gives the practical basis for ultrashort laser pulse processing in possible future applications.

Introduction

I

1 Need for advanced use of ultrashort laser pulses

The generation and manipulation of ultrashort laser pulses play a central role in modern optics, spectroscopy, and nonlinear optical applications. When femtosecond laser pulses propagate through optical components, their temporal characteristics are strongly influenced by the wavelength-dependent refractive index of the materials involved. This effect, known as material dispersion, leads to temporal pulse broadening and phase distortions, which must be carefully controlled in high-precision experiments.

In many practical optical systems, dispersive elements such as laser crystals, lenses, optical windows, and modulators introduce positive dispersion, resulting in an increased pulse duration without a significant reduction in spectral bandwidth. To recover the shortest possible pulse duration at the point of application, dispersion compensation techniques are required.

The aim of this laboratory experiment is to study the physical origin of material dispersion and to demonstrate its compensation using a prism-based pulse compressor. By adjusting the geometry of a two-prism compressor, negative dispersion can be introduced to counterbalance the positive dispersion accumulated in optical components. The effectiveness of the compensation is evaluated by measuring the temporal pulse width using an autocorrelator and comparing the experimental results with theoretical expectations.

2 Laser Safety Considerations

Strict adherence to laser safety regulations is essential during the experiment. The lasers used typically operate in the wavelength range of 760–800 nm, where the human eye sensitivity is very low. This can create the false impression that the emitted optical power is small. In reality, the measured laser powers are significantly higher than those of commonly used He–Ne laboratory lasers.

Furthermore, the lasers operate in an ultrashort pulse regime, which means that ex-

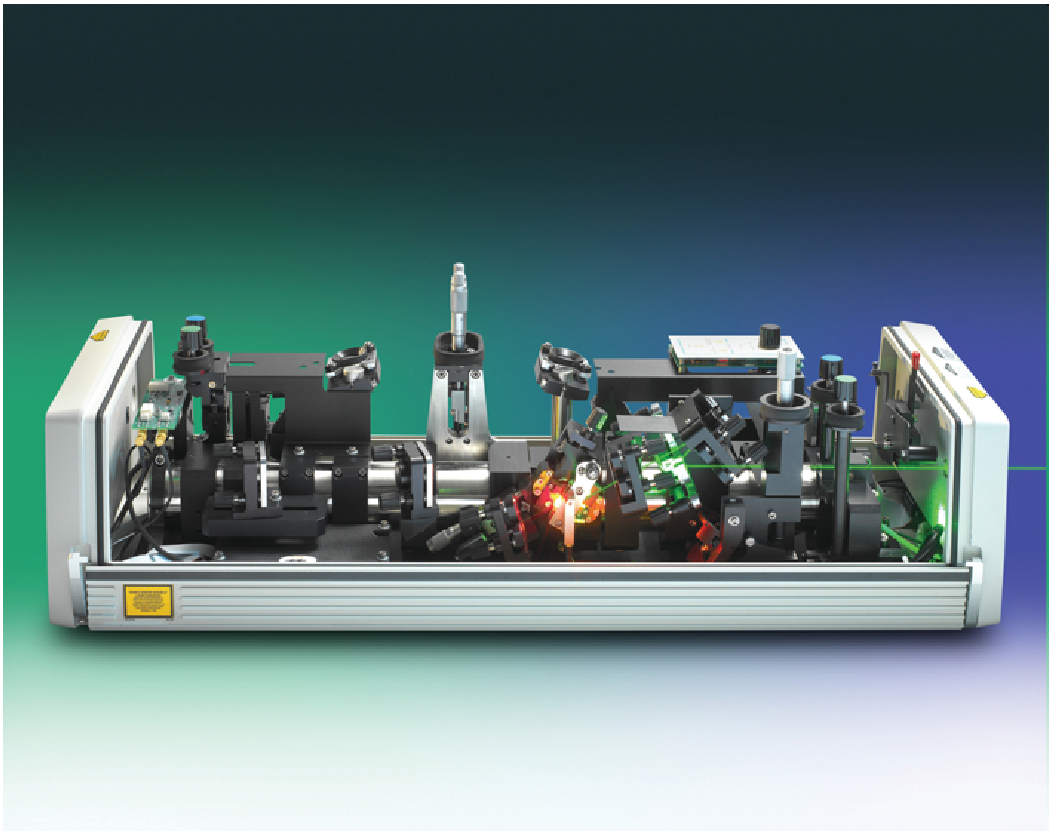

Figure I.1.1. **Ultrashort pulsed laser.** The laser uses an amplifier crystal, T.sapphire, which has an extra high bandwidth of more than 400 nm optical wavelength, between 650 nm and 1100 nm, in the red and near infrared regime of the spectrum. It is pumped by a 5.2 W frequency doubled Nd:YLF laser, at 532 nm wavelength. The pulses generated in the laser are transfrom limited, when quit the laser due to the built in prism compressor tremely high peak intensities are present even at moderate average power levels.

Theoretical Background

II

3 Generation and measurement of ultrashort pulses

3.1 Ultrashort Laser Pulses

The investigation of rapidly evolving physical processes is a fundamental objective in many areas of natural science. Modern laser systems are capable of generating pulses with durations of only a few optical cycles. In some cases, even shorter pulses can be produced using advanced techniques.

A fundamental challenge in measuring such ultrashort laser pulses is that the physical process used for the measurement must be faster than the pulse itself. Conventional detector-based techniques therefore cannot directly measure ultrashort pulses. When a fast photodetector and oscilloscope are used, the measured signal width is determined by the response time of the detection system rather than by the actual pulse duration. Typically, such measurements yield apparent pulse widths on the order of nanoseconds, even though the true pulse duration may be in the picosecond or femtosecond range.

3.2 Mode Locking

Ultrashort pulses are generated using a technique known as mode locking. Mode locking requires a laser resonator and gain medium capable of supporting a large number of longitudinal modes simultaneously.

During mode locking, the phases of the longitudinal modes are synchronized, meaning that the phase differences between the different frequency components are fixed. Mathematically, it can be shown that the superposition of many sinusoidal waves with fixed phase relationships results in the formation of short pulses.

A large number of longitudinal modes also implies that the laser emits radiation over a broad spectral bandwidth. From this spectral width, an estimate can be made of the shortest possible pulse duration assuming perfect phase synchronization. This theoretical minimum pulse duration is known as the transform-limited pulse duration.

3.3 Transform-Limited Pulses and Time–Bandwidth Product

The transform-limited pulse duration depends on assumptions regarding the spectral and temporal shape of the pulse, which are generally not known exactly in practice.

For different pulse shapes, a theoretical relationship exists between the spectral bandwidth and the minimum achievable pulse duration. This relationship is expressed by the time–bandwidth product:

$$\Delta t \cdot \Delta f = \text{constant} \tag{II.3.1}$$

Typical values for different pulse shapes are:

- Gaussian pulse: $\Delta t \Delta f = 0.441$
- Sech^2 pulse: $\Delta t \Delta f = 0.315$
- Lorentzian pulse: $\Delta t \Delta f = 0.142$

Since these pulse shapes are often very similar, their autocorrelation traces are also nearly indistinguishable. Therefore, autocorrelation measurements alone cannot uniquely determine the exact pulse shape.

3.4 Autocorrelation Function

The autocorrelation function is defined as:

$$ACF(\tau) = \int_{-\infty}^{\infty} x(t)\, x(t+\tau)\, dt \tag{II.3.2}$$

In optical autocorrelation measurements, the function $x(t)$ may represent either the electric field or the intensity of the pulse. When field autocorrelation is calculated, the Fourier transform of the optical spectrum is obtained, forming the basis of Fourier-transform spectroscopy.

4 Ultrashort pulse propagation through optical elements

4.1 Material Dispersion

In optical materials, the refractive index depends on the wavelength of the propagating light. For most transparent materials used in optics, this dependence is such that the refractive index decreases with increasing wavelength, a behavior referred to as normal (positive) material dispersion. As a consequence, different spectral components of a short laser pulse propagate with different phase velocities.

For ultrashort pulses, which inherently possess a broad spectral bandwidth, this wavelength-dependent phase accumulation leads to temporal broadening and distortion of the pulse. The effect becomes increasingly significant as the pulse duration decreases and the optical path length through dispersive media increases.

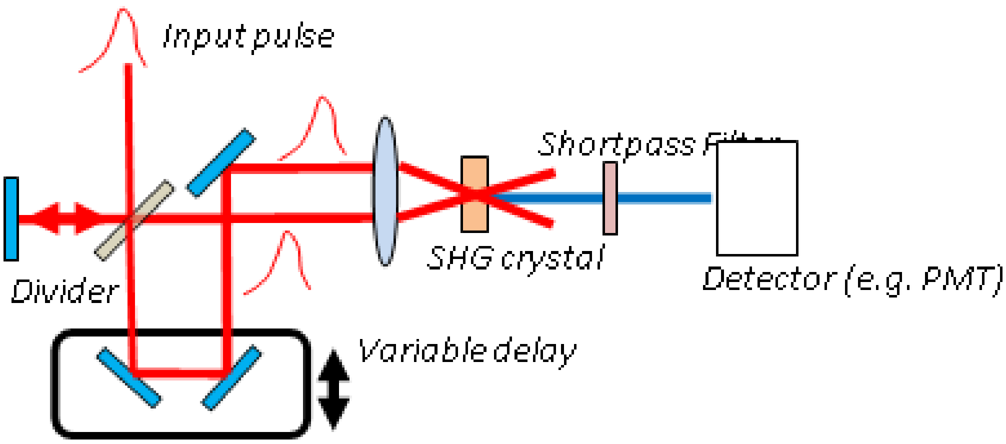

Figure II.3.1. **Autocorrelator.** In this method, the pulse is correlated with a delayed replica of itself, and the pulse duration is estimated from the resulting autocorrelation trace. An optical autocorrelator typically operates by splitting the incoming laser pulse into two beams using a beam splitter. A variable optical path difference is introduced between the two beams, creating a controllable temporal delay. After recombination, the beams are focused into a nonlinear crystal, where second-harmonic generation (SHG) occurs. The intensity of the generated second-harmonic signal is detected as a function of delay. This technique enables the measurement of both picosecond and femtosecond pulses, even at high repetition rates.

4.2 Frequency-Dependent Phase Accumulation

The propagation of a laser pulse through an optical system introduces a frequency-dependent phase shift to its spectral components. This phase can be expressed as a Taylor expansion around the central angular frequency ω_0 of the pulse spectrum:

$$\phi(\omega) = \phi_0 + \left.\frac{d\phi}{d\omega}\right|_{\omega_0} (\omega - \omega_0) + \frac{1}{2} \left.\frac{d^2\phi}{d\omega^2}\right|_{\omega_0} (\omega - \omega_0)^2 + \cdots \tag{II.4.1}$$

The zeroth-order term represents a constant phase shift, while the first-order term corresponds to a temporal delay and does not alter the pulse shape. The second-order term, known as group delay dispersion (GDD), leads to temporal pulse broadening and the formation of a frequency chirp. Higher-order terms, such as third-order dispersion (TOD), become relevant for extremely short pulses and cause more complex distortions of the temporal pulse profile.

4.3 Dispersion Compensation

In ultrafast laser systems, dispersion effects must be compensated in order to maintain short pulse durations. Dispersion compensation is achieved by introducing optical elements that impose a frequency-dependent phase shift of opposite sign to that accumulated in the system.

One of the most widely used dispersion compensation schemes is the prism-pair compressor. In such a configuration, the different spectral components of the pulse are spatially separated by the first prism and experience different optical path lengths between the two prisms. Upon recombination at the second prism, a controllable negative dispersion is introduced.

By properly adjusting the distance between the prisms and the position of the retroreflecting mirror, the negative dispersion of the compressor can be tuned to compensate the positive dispersion introduced by optical materials. When the compensation is optimal, the pulse duration reaches its minimum value, approaching the transform-limited pulse width.

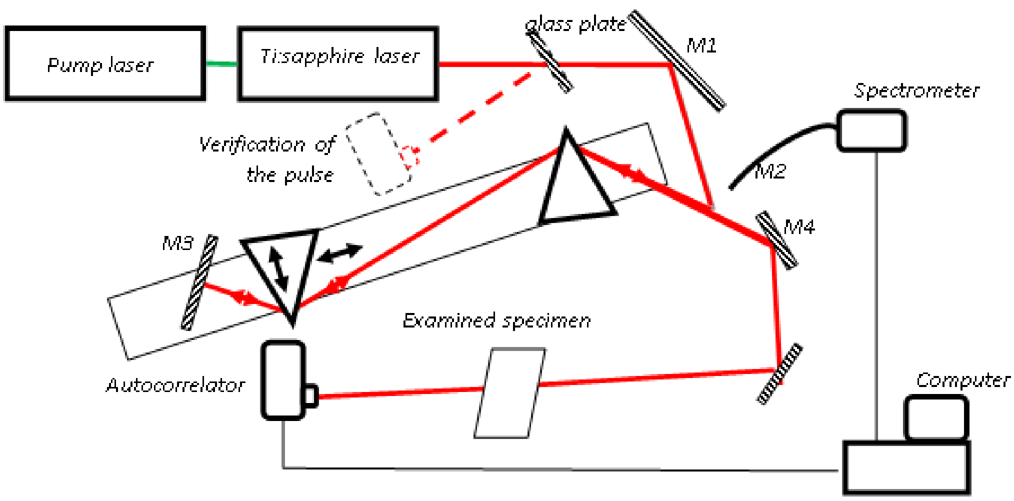

Figure II.4.1. **Prism compressor setup used in the measurement.** Scheme of the prism compressor

In general, a single dispersion-compensating element can compensate only one specific dispersion term, corresponding to a given order in the Taylor expansion of the frequency-dependent phase. Therefore, the overall optical system must be designed by taking all dispersive contributions into account. For femtosecond laser pulses, compensation of the second-order derivative, known as group delay dispersion (GDD), is typically required, while for pulses shorter than approximately 30 fs, the effects of third-order dispersion (TOD) must also be considered.

The most commonly used dispersion-compensating elements are prism pairs. In a prism compressor, material dispersion spatially separates the spectral components of the pulse. Due to the wavelength-dependent optical path length between the prisms, the relative phase of the individual spectral components can be shifted in either a positive or negative frequency direction. In this way, the phase distortions accumulated in the gain medium and on the resonator mirrors can be effectively compensated.

In laser resonators, specially designed multilayer mirrors are also widely employed for dispersion control. In these mirrors, different spectral components of the pulse are reflected from different penetration depths depending on their frequency, resulting in a frequency-dependent phase shift. The mirrors used in resonators must be designed such that the phase shift they introduce precisely compensates the frequency-selective phase accumulation occurring in the gain medium.

With proper optical design, the application of anti-reflection coatings, and operation at the Brewster angle, both prisms and mirrors can exhibit low optical losses. As a result, they can be integrated into laser resonators without significantly reducing the extractable pulse energy. In contrast, diffraction gratings may also be used for dispersion compensation, but typically only outside the resonator or in high-gain fiber laser systems, since their optical losses are considerably higher.

Nonlinear optical effects play a crucial role in shaping the temporal and spatial characteristics of ultrashort laser pulses. These effects arise from the intensity dependence of the refractive index in various materials within the resonator and the propagation medium, leading to intensity-dependent phase shifts experienced by different parts of the pulse. The most common nonlinear effect is the Kerr effect, in which the refractive index n depends linearly on the optical intensity I:

$$n = n_0 + n_2 I. \tag{II.4.2}$$

As a consequence of nonlinear effects, laser pulses become distorted both transversely in space and temporally. The high-intensity regions of the pulse, corresponding to its spatial and temporal peak, experience a larger phase delay than the pulse wings. The resulting temporal distortion is known as self-phase modulation (SPM).

The spatial effect of the Kerr nonlinearity forms the basis of one of the most successful passive mode-locking techniques, Kerr-lens mode locking (KLM). Furthermore, the combined temporal effects of dispersion and nonlinearity can lead to soliton formation, in which the two effects mutually compensate each other. This mechanism is generally more complex, as both second-order and higher-order dispersion terms can interact with nonlinear effects.

A fundamental soliton is formed when the second-order dispersion (GDD) and the Kerr effect have opposite signs and comparable magnitudes over one round trip in the resonator. In this case, the pulse remains stable both spatially and temporally, preserving its shape and duration during propagation, while accumulating only a nonlinear phase shift. In the presence of higher-order dispersion, higher-order solitons may form, for which the pulse shape and duration vary periodically along the propagation direction.

The generation of pulses shorter than approximately 100 fs is only possible through soliton formation, as optical switching elements with sufficiently fast response times do not exist, even in passive implementations.

Solitons formed outside the laser resonator are particularly important for long-distance optical communication, primarily in optical fibers. Although soliton formation in free space is highly unlikely, it has been observed in exceptional cases. Optical fibers provide favorable conditions for self-phase modulation, and their intrinsic dispersion can be tailored through material composition and doping.

4.4 Dispersion Compensation of External Optical Elements

External optical components, such as acousto-optic crystals or lenses, introduce additional dispersion that stretches the pulse duration without significantly reducing its spectral bandwidth. This dispersion must be compensated in such a way that, at the point of application, the original transform-limited pulse corresponding to the available bandwidth is restored. The simplest and experimentally adequate method for this purpose is the use of a two-prism compressor. In this configuration, the laser beam passes through the prism pair twice, and due to the dispersion introduced by the prisms, it experiences a negative group delay dispersion that counteracts the positive dispersion accumulated in the crystals and other optical elements. The compressor is correctly adjusted when the distance between the prisms and the retroreflecting mirror is such that the introduced negative dispersion compensates the total positive dispersion of the system.

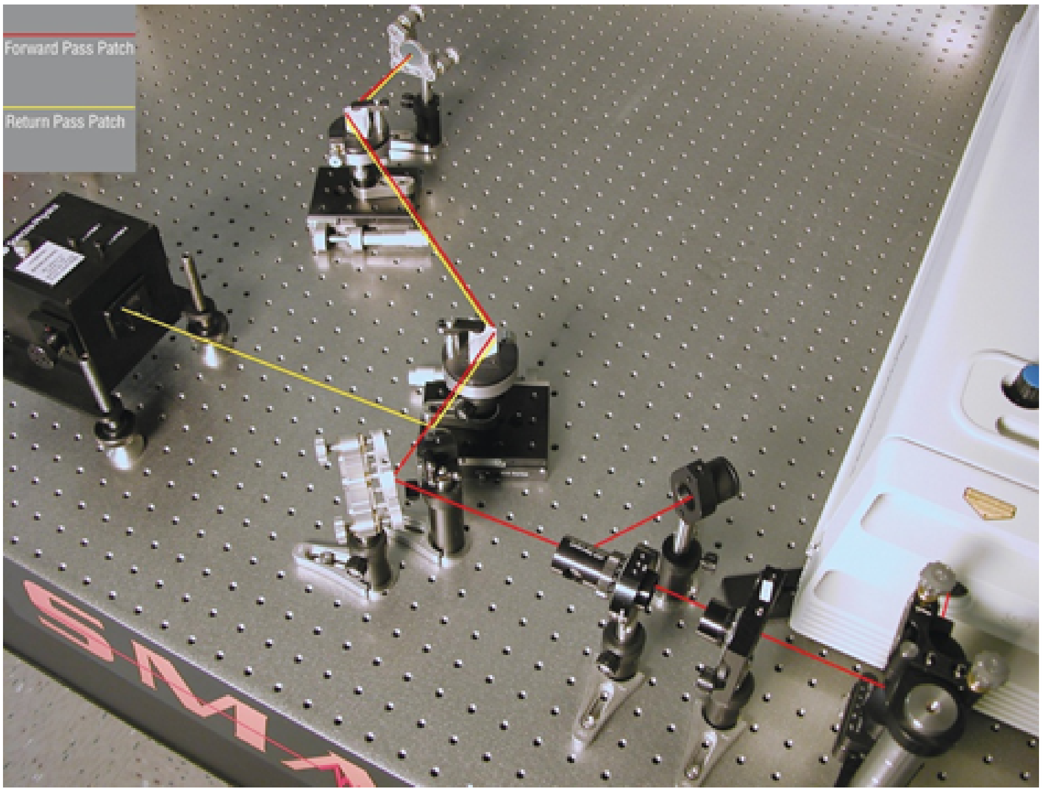

Figure II.4.2. **Two pass prism compressor.** Photograph of a two-pass prism compressor. The angular dispersion and spatial lateral spectral shift after the first propagation through the prism pair is removed during the second, reversed propagation through the pair.Therefore it is very important that the two propagation tracks are strictly parallel

This adjustment must be verified during the measurement and modified if necessary in order to achieve optimal pulse compression.

5 Angular Dispersion and Spectral Interferometry

6 Evaluation of Spectrally Resolved Interferograms

6.1 Principle of Spectral Interferometry

Spectrally resolved interferometry is a high-precision measurement technique suitable for determining the dispersion of optical elements. In this method, an interferometer containing the sample under investigation is illuminated by a broadband light source (e.g., a halogen lamp or a Ti:sapphire laser).

When the reference arm length is properly adjusted, interference fringes appear at the interferometer output. These fringes are spectrally resolved by a spectrometer. By placing a linear detector array or camera in the image plane of the spectrograph, the recorded interferograms can be analyzed numerically. The frequency-dependent intensity distribution at the spectrometer output can be written as

$$I(\omega) = I_R(\omega) + I_T(\omega) + 2\sqrt{I_R(\omega) I_T(\omega)} \cos\left[\Delta\phi(\omega)\right], \tag{II.6.1}$$

where $I_R(\omega)$ and $I_T(\omega)$ are the spectral intensities of the reference and test arms, respectively, and $\Delta\phi(\omega)$ denotes the phase difference between the two interferometer arms.

The frequency-dependent phase term can be expressed as

$$\Delta\phi(\omega) = \phi(\omega) + \omega\tau, \tag{II.6.2}$$

where $\phi(\omega)$ is the spectral phase introduced by the sample and τ is the temporal delay corresponding to the air-path difference between the interferometer arms.

6.2 Constant Phase Point

To determine the extrema of the normalized interferogram, we examine the derivative of the phase term with respect to frequency. An extremum occurs when

$$\frac{d}{d\omega}\Delta\phi(\omega) = 0. \tag{II.6.3}$$

At the frequency satisfying this condition, the so-called constant phase point is formed. Around this frequency, the phase varies only slowly with respect to ω, and the fringe spacing locally becomes larger.

Changing the delay τ shifts the position of the constant phase point along the spectrum. Substituting Eq. (II.6.2) into Eq. (II.6.3), we obtain

$$\frac{d\phi(\omega)}{d\omega} + \tau = 0. \tag{II.6.4}$$

Since $d\phi/d\omega$ corresponds to the group delay introduced by the sample, the position of the constant phase point provides information about the sign and magnitude of the group delay.

6.3 Minimum–Maximum Evaluation Method

The extrema of the interferogram occur when

$$\sin\left[\Delta\phi(\omega)\right] = 0, \tag{II.6.5}$$

which corresponds to phase values that are integer multiples of π:

$$\Delta\phi(\omega_m) = m\pi, \tag{II.6.6}$$

where m is an integer index assigned to successive maxima and minima.

In practice, the measured and normalized interferogram is differentiated numerically, and the zero-crossings of the derivative (sign changes) are identified to locate the extrema frequencies ω_m. By assigning integer indices m to the extrema (with $m = 0$ at the constant phase point if present within the measured spectral range), the spectral phase values corresponding to each extremum can be reconstructed.

Plotting $m\pi$ as a function of ω_m yields a discrete representation of the spectral phase function. By fitting a polynomial to this curve, one effectively fits the Taylor expansion of the phase:

$$\phi(\omega) = \phi_0 + \phi_1(\omega - \omega_0) + \frac{1}{2}\phi_2(\omega - \omega_0)^2 + \frac{1}{6}\phi_3(\omega - \omega_0)^3 + \dots \tag{II.6.7}$$

The coefficients correspond to:

- ϕ_1 – group delay,
- ϕ_2 – group delay dispersion (GDD),

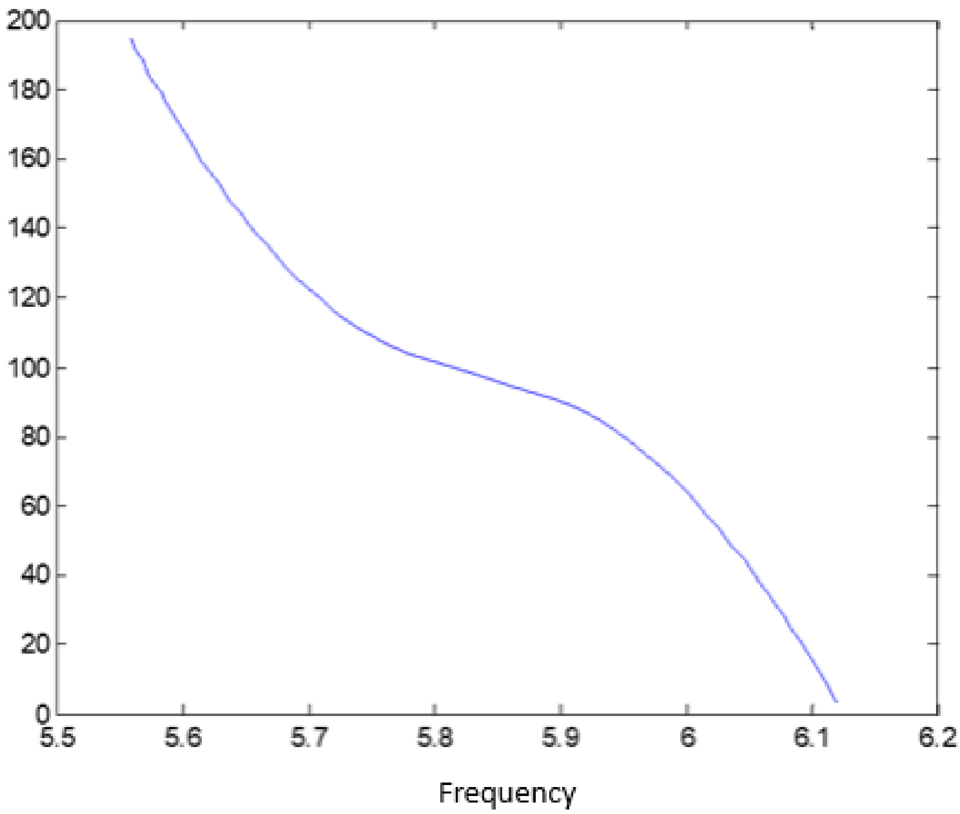

Figure II.6.1. **Phase as a function of frequency.** Phase values obtained from the minimum-maximum fitting

- ϕ_3 – third-order dispersion (TOD).

If a second-order polynomial is fitted, only the GDD is obtained. Including a third-order term allows extraction of the TOD as well. The order of the polynomial that can be reliably fitted depends on the number of measured extrema and the noise level of the interferogram. A broader spectrum yields a larger number of extrema and therefore allows higher-order dispersion terms to be determined more accurately.

The precision of the method can be improved by increasing the delay between the interferometer arms, which results in a denser fringe pattern. A broader spectral bandwidth also increases the measurement accuracy.

A limitation of the minimum–maximum method is the potentially laborious identification of extrema, especially in the presence of noise.

6.4 Angular Dispersion

One possible source of pulse distortion is the material dispersion of optical components, whose temporal effects can be efficiently compensated using pulse compressors. However, pulse distortion may also occur due to angular dispersion introduced by optical elements.

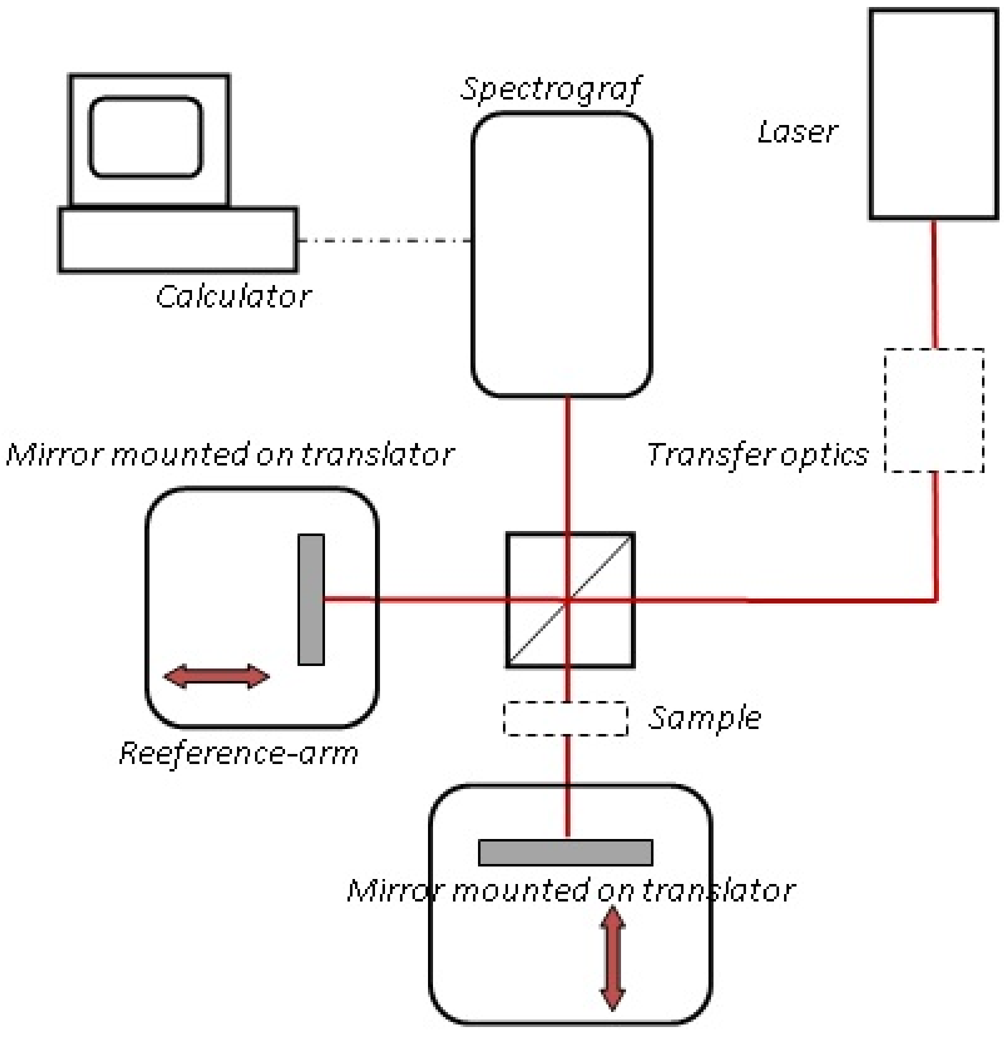

Figure II.6.2. **Interferometer setup.** A typical Michelson interferometer used for measuring spectrally resolved interferograms of ultrashort pulses

Angular dispersion arises when the different spectral components of a pulse propagate in different directions after passing through an optical element. Typical elements that introduce angular dispersion include prisms, diffraction gratings, acousto-optic deflectors, as well as pulse stretchers or compressors that are not perfectly aligned.

6.5 Physical Interpretation of Angular Dispersion

In the case of material dispersion, pulse broadening occurs because monochromatic components of different wavelengths propagate in the same direction but accumulate different phase shifts due to wavelength-dependent phase velocities.

A phase difference may also arise when the spectral components propagate at the same speed but in different directions. This situation occurs when a pulse passes through an element with frequency-dependent refraction (such as a prism) or reflection angle (such as a diffraction grating), and subsequently propagates in a low-dispersion medium such as air.

Outside dispersive media, angular dispersion can exist without temporal chirp, while inside materials both effects may occur simultaneously.

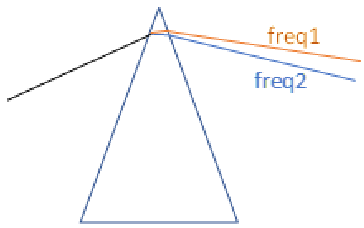

Figure II.6.3. **Concept of angular dispersion created by prism.** The different colors represent different frequency components of a fs pulsed. In reality their frequency doesn't differ enough to cause real color difference, since the full bandwidth is of about 20 nm

6.6 Definitions of Angular Dispersion

In practice, two different definitions of angular dispersion are used.

The first definition describes angular dispersion as the frequency dependence of the propagation direction of the light beam:

$$\frac{d\theta}{d\omega}, \tag{II.6.8}$$

which is referred to as propagation-direction angular dispersion.

The second definition describes the frequency dependence of the angle between the phase fronts of monochromatic components:

$$\frac{d\phi}{d\omega}, \tag{II.6.9}$$

known as phase-front angular dispersion.

For plane waves, these two definitions yield identical results. However, for beams with curved phase fronts, such as Gaussian beams, the two quantities differ. The phase-front angular dispersion manifests as pulse-front tilt, while propagation-direction angular dispersion results in combined spatial and temporal phase modulation.

6.7 Measurement of Dispersion Using Spectral Interferometry

Spectrally resolved interferometry is a high-precision measurement technique that can be used to determine the dispersion of optical components. In this method, an interferometer

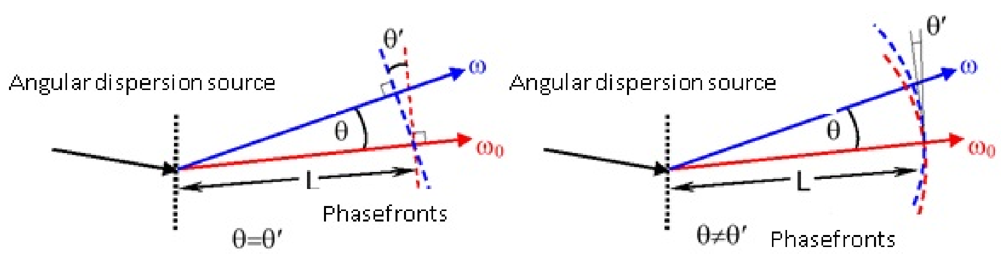

Figure II.6.4. **Ultrashort pulsed laser.** The two definitions of the angular dispersion yield different behavior of the spatial dispersion in the case of plane waves and Gaussian beams containing the sample under investigation is illuminated with a broadband light source, such as a halogen lamp or a Ti:sapphire laser.

Interference fringes appearing at the interferometer output are spectrally resolved using a spectrometer. By placing a linear detector array at the spectrometer image plane, the recorded interferograms can be analyzed computationally to extract dispersion information. The frequency-dependent intensity distribution of the recorded interferogram can be expressed as:

$$I(\omega) = I_1(\omega) + I_2(\omega) + 2\sqrt{I_1(\omega)I_2(\omega)}\cos\left[\Delta\phi(\omega)\right], \tag{II.6.10}$$

where $\Delta\phi(\omega)$ denotes the frequency-dependent phase difference between the interferometer arms.

6.8 Measurement of Propagation-Direction Angular Dispersion

The propagation-direction angular dispersion is measured using a spectrograph. The investigated beam is focused onto the entrance slit of the spectrograph using a focusing element such as an achromatic lens or a concave mirror.

Due to angular dispersion, the spectral components of the pulse propagate in slightly different directions. Consequently, they are focused onto different vertical positions at the slit plane. Inside the spectrograph, the slit image is projected onto a CCD camera, while the diffraction grating introduces wavelength-dependent horizontal dispersion.

As a result, the recorded spectrum appears tilted on the CCD detector. The tilt angle is proportional to both the angular dispersion and the focal length of the focusing lens.

By selecting appropriate spectrograph parameters, angular dispersion can be measured with an accuracy better than 0.2 rad/nm. This method enables real-time measurements and

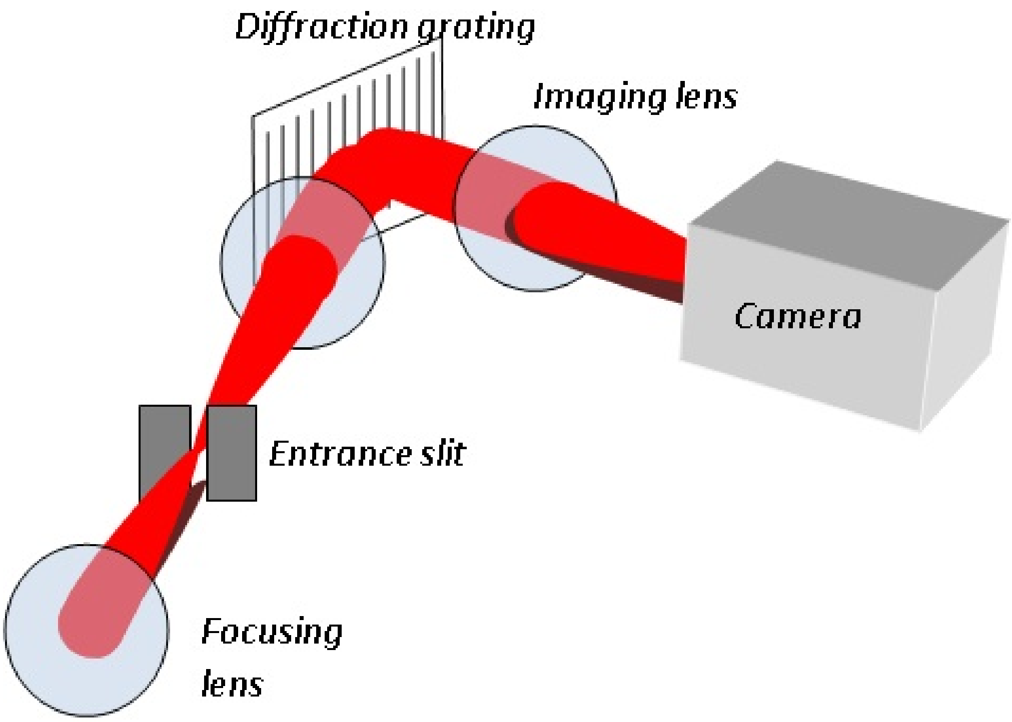

Figure II.6.5. Ultrashort pulsed laser. The angularly dispersed beam is simply focused to the spectrometer input slit by a 200 mm focal length lens is therefore particularly suitable for alignment of pulse stretchers and compressors. During the measurement, the tilted spectral image is recorded with the camera. The saved image is analyzed using numerical software such as MATLAB or Excel. Vertical intensity profiles are extracted at different wavelengths, and the vertical positions of intensity maxima are determined.

Considering the pixel size and the magnification of the spectrograph imaging system, the ratio of vertical displacement to wavelength change can be calculated. Since the angular deviations are small, the angular dispersion can be approximated by:

$$\frac{d\theta}{d\lambda} \approx \frac{y}{f\,\Delta\lambda}, \tag{II.6.11}$$

where f denotes the focal length of the focusing lens. The software - Fringer - associated to the spectrometer also measures this angular dispersion. This can serve as control for the evaluation with other software, as listed between the measurement tasks.

6.9 Measurement of Phase-Front Angular Dispersion

Phase-front angular dispersion is measured using the same spectrograph combined with a Mach–Zehnder interferometer. The interferometer is adjusted such that one beam undergoes an additional reflection compared to the other, resulting in overlapping beams at the spectrograph entrance slit.

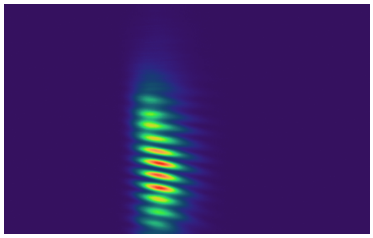

Figure II.6.6. Interference pattern without angular dispersion. The interference stripes are parallel

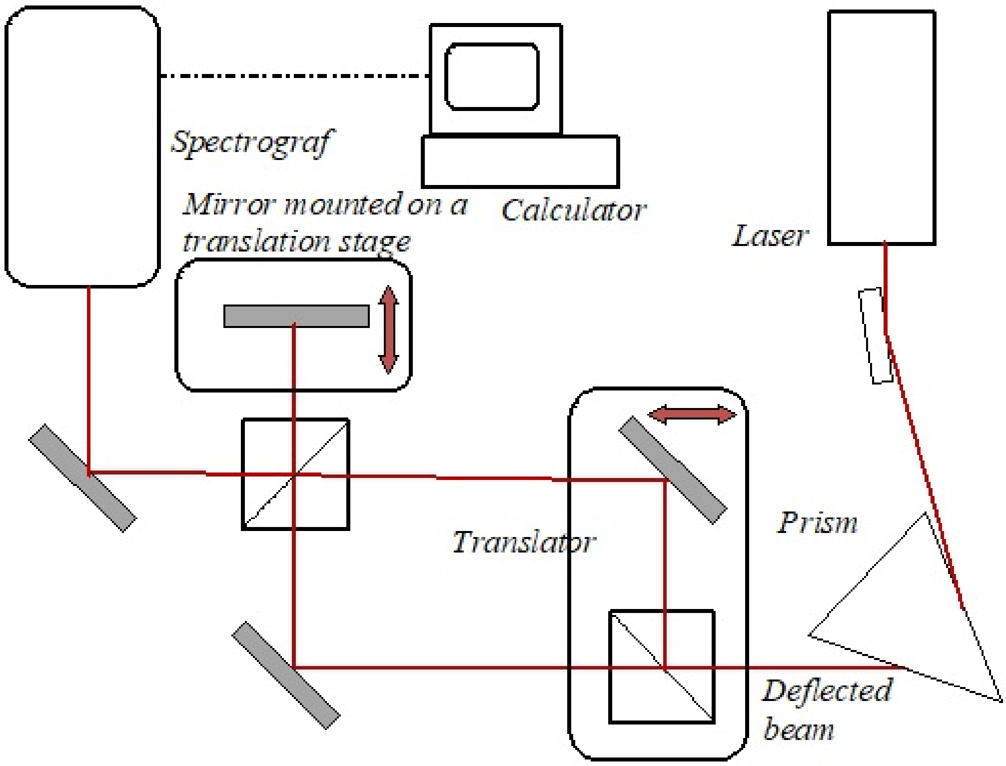

Figure II.6.8. Mach-Zehnder interferometer combined with a spectrometer. The interferometer produces interference between laterally spatially dispersed beams. One of the arms contains one reflection more than the other, hence different frequency beam portions are overlapped, when lateral spatial dispersion is in the beams

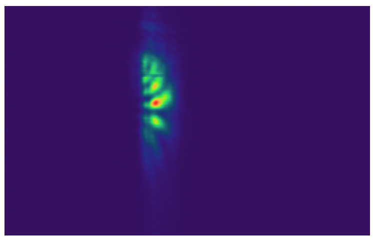

Figure II.6.7. Interference pattern with strong angular dispersion. The interference stripes are strongly curved

By slightly tilting one of the interferometer mirrors, interference fringes perpendicular to the slit direction are generated. If angular dispersion is present, the interference fringes become curved in the spectrally resolved interferogram.

Quantitative determination of angular dispersion is achieved by analyzing the wavelength dependence of the fringe spacing. If the spectral components are parallel, the fringe period varies linearly with wavelength. Deviations from linearity indicate the presence of angular dispersion.

7 Investigation of Ultrashort Laser Pulses Using an Autocorrelator

7.1 Preliminary Verification

First, verify using a fast photodetector that the pulse duration is shorter than the temporal resolution limit of the available detector. This confirms that conventional electronic measurement techniques are insufficient for direct pulse duration determination.

7.2 Laser Initialization and Spectral Characterization

Start the femtosecond laser system. If a tunable laser is used (e.g., a Ti:sapphire laser), select the desired operating wavelength using the spectrometer. Establish communication between the spectrometer and the computer, and if required, between the laser and the control software.

Measure the output optical power and record the spectral profile of the laser emission. Use neutral density filters or beam sampling plates if necessary to prevent detector saturation.

7.3 Determination of Transform-Limited Pulse Duration

Using the measured spectral bandwidth, estimate the transform-limited pulse duration assuming different pulse shapes, such as Gaussian and Sech^2 profiles.

7.4 Autocorrelator Setup and Direct Pulse Measurement

Determine the maximum permissible input power of the autocorrelator based on its sensitivity specifications. Adjust the input power accordingly using beam samplers or attenuators.

Align and calibrate the autocorrelator, and measure the pulse duration of the beam directly exiting the laser, ensuring that the beam propagates through as few optical elements as possible.

Determine the pulse duration under different pulse-shape assumptions by:

- direct readout from the autocorrelator display,
- numerical evaluation using computational tools such as MATLAB.

Compare the measured pulse duration with the previously calculated transform-limited value.

Measurement Tasks

III

8 Dispersion Compensation Using a Prism Compressor

8.1 Experimental Equipment

The following equipment is used during the experiment:

- Femtosecond laser system (average power $\sim 400\,\mathrm{mW}$, pulse duration $\sim 50\,\mathrm{fs}$, spectral FWHM $\sim 12\,\mathrm{nm}$)
- Avantes spectrometer operating in the 500–1000 nm wavelength range
- Coherent optical power meter
- APE autocorrelator
- glass plate for outcoupling part of the power for measurement
- Prism compressor aligned with the beam path:
  - first prism on translation stage
  - second prism on translational and rotational stage
- mounted mirror for outcoupling after the compressor
- dispersive samples of different materials
- computer software to evaluate autocorrelator traces

8.2 Measurement Procedure

Objective of the Measurement The aim of this task is to gain practical experience in the positioning and operation of an optical autocorrelator, as well as in the routine use of a spectrometer and optical power meter. The students perform autocorrelation measurements of ultrashort laser pulses and apply various experimental techniques necessary for the successful execution of the measurement.

An additional objective is the numerical calculation of autocorrelation functions using computer software and the critical evaluation of the obtained experimental results.

1. Laser startup and mode-locking verification.

Switch on the Ti:sapphire laser by tuning the pump laser to maximum power. Verify that the output power exceeds 400 mW using the power meter. Turn on pulsing by pushing the enable button of the acousto-optic modelocker.

Start the spectrometer software (AVASOFT) and display it in full-screen mode. Direct the spectrometer fiber input toward a visible scattered spot from a mirror and observe the spectrum on the screen. If necessary, optimize the output power and spectral shape by adjusting the end mirrors and intracavity prisms while continuously monitoring both the spectrum and the output power. At the final stage, initiate stable pulsed operation by rapidly adjusting the prism micrometer.

2. Spectral measurement.

Use the spectrometer to confirm that the laser operates in mode-locked pulse regime. The laser is considered to operate properly in pulsed mode if the spectral bandwidth exceeds 10 nm. With the cursor measure the wavelength full width half maximum of the spectrum. Save the recorded spectrum for later evaluation.

3. Initial pulse duration measurement.

Extract a small fraction of the laser beam using a glass plate and direct the reflected beam (which does not pass through the glass and is therefore approximately transform-limited) into the autocorrelator.

Align the autocorrelator using the crosshair reference and adjust its position until an optimal signal amplitude is obtained. Ensure that the autocorrelator housing is perpendicular to the incident beam.

Measure and record the temporal pulse shape and pulse duration of the laser output.

4. Prism compressor alignment.

Using the appropriate mirror, guide the laser beam into the prism compressor. Adjust the prisms to operate at Brewster angle such that the intensity of the beam reflected from the first prism surface is minimized. This can be checked visually on a screen or more precisely with the power meter.

The rear mirror of the compressor vertically displaces the incoming and outgoing beams. Ensure that both beams pass entirely through the prisms and that the outgoing beam exits directly above the incoming beam. The exiting beam must pass above the coupling mirror without obstruction.

5. Pulse measurement after compressor.

Use a glass plate to sample the compressed beam and direct the reflected portion into the autocorrelator. Measure both the pulse duration and the corresponding spectrum.

6. Compensation of additional dispersion (glass plate).

Insert a prepared 1 cm thick glass plate between the compressor output mirror and the beam-sampling plate.

Realign the autocorrelator to obtain maximum signal. If the measured pulse duration exceeds 300 fs, replace the plate with a thicker glass block.

Using the translation stage, adjust the position of the second prism in the compressor to minimize the pulse duration measured by the autocorrelator. Ensure that the outgoing beam remains directly above the incoming beam and that the beam is not clipped by prism edges. This can be achieved by fine adjustment of the second prism in both translational directions.

Record:

- the autocorrelation trace,
- the pulse duration assuming a Sech^2 pulse shape,
- the distance between the two prisms,
- the distance between the second prism and the retroreflecting mirror.

7. Measurement with glass cube.

Replace the thin glass plate with a glass cube. Adjust the compressor again to minimize the pulse duration measured by the autocorrelator (realign the autocorrelator if necessary).

Record:

- the autocorrelation signal,
- the emission spectrum,
- the pulse duration (Sech^2 assumption),
- prism separation distances with a tape measure.

8. Comparison and analysis.

Compare the measured pulse durations for the beam transmitted through the glass plate and the glass cube. Compare the corresponding prism separations in the compressor with the physical lengths of the glass samples along the propagation direction.

If the ratios of glass thickness and compressor length adjustment differ, provide a physical explanation for the discrepancy.

9. Compensation of dispersion introduced by an acousto-optic element.

Insert an acousto-optic filter between the compressor output mirror and the beam-sampling plate.

Adjust the prism separation and relative prism positions to minimize the pulse duration measured by the autocorrelator. If the travel range of the second prism is insufficient, release the mechanical mounting screws and reposition the second prism assembly relative to the first prism, then secure it again.

Record the autocorrelation trace, pulse duration (Sech^2), prism separation, and the distance between the second prism and the retroreflecting mirror.

9 Experimental Setup and Measurement Procedure for spectral interferometry

9.1 Experimental Equipment

The following equipment is used during the experiment:

- He-Ne laser for preliminary adjustment of the model interferometer
- Model interferometer for adjustment training
- Femtosecond laser system (average power $\sim 800\,\mathrm{mW}$, pulse duration $\sim 50\,\mathrm{fs}$, spectral FWHM $\sim 16\,\mathrm{nm}$)
- Avantes spectrometer operating in the 500–1000 nm wavelength range
- Optical power meter
- Two interferometric setups mounted on linear translation stages:
  - Michelson interferometer
  - Mach–Zehnder-type interferometer (one arm containing one additional reflection)
- Each interferometer includes:
  - at least two angle-adjustable mirrors,
  - at least one beam splitter cube,
  - input and output apertures

- All mirrors are mounted on linear translation stages enabling displacement perpendicular to the mirror surface
- BK7 glass cube for dispersion measurements
- glass prism for generating angular dispersion
- High-resolution spectrograph with CCD camera for interferogram acquisition
- Computer running the FRINGER evaluation software
- MATLAB or Excel for numerical analysis

9.2 Spectral Interferometry Measurement

1. Adjusting the model interferometer Switch on the He-Ne laser and verify the beam path through the beam expander. Verify that the beam propagates parallel to the optical table before the interferometer using a mounted variable aperture. Adjust the beam direction and the mirrors of the interferometer until the beams at the output are completely overlapping. Observe the interference pattern on a screen and the change of the stripes when the reflection angle of one of the mirrors is adjusted.

2. fs laser characterization. Switch on the laser if it is not already running and verify both the output power and the emission spectrum. Estimate the spectral bandwidth (FWHM). If the bandwidth is below 10 nm, adjust the laser (with the instructor's assistance) to ensure proper mode-locked operation.

3. Michelson interferometer alignment. Set up the Michelson interferometer such that the two output beams overlap along the entire propagation path from the beam splitter to the spectrometer. Adjust the arm lengths to be approximately equal using a measuring tape.

4. Spectrometer alignment. Insert the spectrometer and align the beams onto the entrance slit. Observe the spatial overlap of the two beams in the non-dispersed direction on the CCD camera. Once interference fringes appear, optimize fringe sharpness and contrast using mirror adjustments and fine (micrometer-resolution) linear translation stages. A single line distance corresponds to 0.2 micrometers on the finer translation stage.

5. Interferogram acquisition. Record interferograms at several different arm-length differences. Additionally, record the individual intensity spectra of the reference and sample arms (without interference), which are required for normalization. With the light blocked, measure the background signal and enable automatic background subtraction.

6. Measurement with dispersive sample. Insert the BK7 glass cube into the sample arm and adjust the interferometer to restore equal arm lengths. Record interferograms at several delays. Again, measure the individual spectra of both interferometer arms and record the background before starting the acquisition.

7. Dispersion extraction. If the data quality permits, register interferogram for both the empty interferometer and the configuration including the dispersive sample. Determine the group delay dispersion (GDD) introduced by the sample from the shift of the interferogram due to the sample, and if the signal-to-noise ratio allows, also extract the third-order dispersion (TOD).

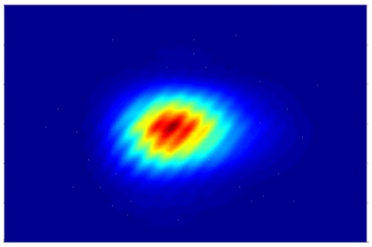

Figure III.9.1. **Typical interference pattern.** The interference stripes are shifted with the ssample introduced into the beam path

9.3 Angular Dispersion Measurement

8. Generation of angular dispersion. Guide the laser beam through the prism ensuring incidence at Brewster angle at the first prism face. Direct the diffracted beam through the Mach–Zehnder interferometer. At the interferometer output aperture, ensure spatial overlap of the beams.

9. Adjustment of the spectrometer and interferometer Position the spectrometer after the interferometer, parallel to the interferometer axis. Insert the mirror-based beam steering and polarization control system between the interferometer output and the spectrometer entrance slit. Adjust mirror angles and heights to achieve proper alignment and visible interference fringes on the CCD camera.

Adjust the optical path difference using the linear translation stage until interference fringes appear. Ensure that the optical path between the interferometer and the spectrometer is at least 300 mm.

10. Propagation-direction angular dispersion measurement. Block one interferometer arm. Insert a focusing lens ($f = 100\,\mathrm{mm}$ or $f = 200\,\mathrm{mm}$) before the spectrometer to focus the beam onto the entrance slit.

Record the tilted spectrum on the CCD camera. Fit a straight line to the tilted spectral trace according to the theoretical relation:

$$\frac{d\theta}{d\lambda} \approx \frac{y}{f\Delta\lambda}, \tag{III.9.1}$$

and determine the value of propagation-direction angular dispersion.

11. Software comparison. Compare the manually calculated angular dispersion with the corresponding value obtained from the FRINGER software fitting procedure.

12. Interferometric verification. Remove the beam block and allow both beams to propagate through the interferometer and spectrometer. Record interferograms at several arm-length differences and evaluate them using FRINGER. Compare the results with those obtained from the direct angular-dispersion measurement.

9.4 Control Questions

1. What is meant by the term *spectral interferometry*?
2. What ensures temporal coherence necessary for the formation of interference fringes?
3. What determines the position of a constant-phase point in the spectrum?
4. How does the minimum–maximum method for interferogram evaluation work?
5. Sketch the main difference between interferometric setups used to measure angular dispersion and those used to measure temporal pulse broadening (material dispersion).